

Build brief — the spine

DR–B periods & L -functions note

Contribution, stated narrowly

The object is a finite, regulator-free, two-segment contour functional with a Hurwitz-zeta T -kernel that delivers period polynomials and L -values from one construction, recovers BFK on specialization, and extends — within a homotopy class — to interior-pole meromorphic forms. The framework is Brown’s (arXiv:1710.07912): finite base point, τ_0 -independent cohomology class, T -gauge. The contribution is the explicit closed kernel and the unified finite realization, *not* new objects — the L -values are BFK’s, the periods are Brown’s quasi-periods made closed-form.

Language: a manifestly convergent finite formula that *equals* the regularized value is a certificate of canonicity — scheme-, continuation-, and cusp-choice-independent — not a correction to results that were already rigorous. Lean on *canonical* / *certified*, never on *correct*.

The spine

Lead with the construction question, so the kernel is an *answer*:

Periods are a cocycle. How do you build one — parabolic — for a modular form from a generic finite base point?

Precision 1: the gate is the constant term, not the pole structure. The parabolic gauge (killing C_T) exists iff $c_f(0) = 0$; the principal part never obstructs it, interior poles included. “Generic pole structure” is the right phrase and a selling point — arbitrary poles are fine. But “generic form” at the cocycle step is too generous: $c_f(0) \neq 0$ (an Eisenstein series) cannot be made parabolic and yields the rational/Eisenstein piece. Clean statement: *any form with vanishing constant term, regardless of pole structure, admits the parabolic gauge*. The gate is one Fourier coefficient.

Precision 2: the two legs have different gates — don’t bundle them. The period-polynomial leg needs the parabolic gauge ($c_f(0) = 0$, i.e. $f \in S_k^!$). The L -function leg does not: the two-segment functional is defined on all of $M_k^!$, constant term allowed, with the expected poles of Λ coming from $c_f(0)$ as classically. The arc *widens* as it goes; let that be visible.

Presentation: define and verify; derive in a footnote. Do not walk the reader through the variation-in- T operator, the linear algebra, and the Bernoulli combinatorics. Present $\tilde{k}_T$ as a definition, apply δ_T , and show the parabolic obstruction vanishes (for f with $c_f(0) = 0$) at all s — a two-line check. Push the provenance to a footnote: (1) construct the variation-in- T operator δ_T ; (2) set up the coboundary linear algebra; (3) read off the combinatorics; (4) recognize them as Bernoulli, i.e. special values of $\zeta(s, x)$ at integer s ; (5) note that this $\zeta(s, x)$ is already entire in x and meromorphic in s , so the inversion is a *function* identity with the integers as one slice. The footnote says the integer case *revealed which analytic object was operating* — not that a discrete

identity was analytically continued. The second framing invites a continuation-validity question the construction does not carry.

The four beats (~5 pp with disciplined front/back matter).

1. **Periods are a cocycle; the construction question.** Eichler–Shimura in a paragraph; the task is a parabolic cocycle from a generic finite base point. This is Brown (arXiv:1710.07912) — recall it, with the gate $c_f(0) = 0$ and its earlier form as constant-term subtraction $\tilde{f}|(1-T) = 0$ (arXiv:1202.5802).
2. **Define the Hurwitz T -kernel; verify it inverts δ_T .** $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)}\zeta(s-(k-1), \tau+1)$, implicit in Brown’s P ; the contribution is the explicit all- s form. Verify, don’t derive: $\delta_T \zeta(1-s, \tau+1) = -(\tau+1)^{s-1}$ inverts δ_T uniformly in s , so applying δ_T to the kernel leaves exactly the constant-term piece — which is why the obstruction vanishes *iff* $c_f(0) = 0$. State that gate here even in the short version; the verification is precisely where it appears. On $S_k^!$ this beat delivers the period polynomial $r_f/\omega^\pm$ as the two-segment functional.
3. **Swap the period monomial for $\tau^{s-1} \rightarrow$ the L -function.** Same kernel; the functional is entire in s , and $L^*(f, s)$ is entire on $S_k^!$, with the expected poles otherwise — on $M_k^!$ it is meromorphic, carrying exactly the poles forced by $c_f(0)$, as classical Λ does. (Say “entire on $S_k^!$, expected poles otherwise”; do not call the $M_k^!$ object entire.) The $\tau_0 = i$ specialization recovers BFK with the pair $\Gamma(s, 2\pi n)$ and $i^k \Gamma(k-s, 2\pi n)$ — the $s \leftrightarrow k-s$ reflection making $\Lambda^*(s) = i^k \Lambda^*(k-s)$ manifest; $b = 1$ is the split *height* ($\tau = i$, the S -fixed point), not the s -center. All- s is a *result*: the term-by-term unfolding gives a manifestly meromorphic incomplete-gamma series matching the classical / Apostol / BFK expressions coefficient by coefficient, so there is no continuation step — only the finite principal-part interchange (the Hurwitz-unfolding proposition).
4. **Encapsulation.** One object covers both literature regimes: weak / regular cusps with arbitrary principal part (fully canonical, simply connected), *and* interior-pole meromorphic forms — canonical given a contour, the cross-wall dependence being residues. State the winding prescription explicitly (the symmetric $\frac{1}{2} \sum_{\pm\epsilon}$ of arXiv:1806.09874): the identity is then definitional. The jump at an interior pole is exactly f ’s residue times the finite kernel value there — $\zeta(\sigma, \tau+1)$ is holomorphic throughout $\mathbb{H}$, so the kernel contributes no residue of its own.

References to genuflect toward

BFK (2014); arXiv:1809.02938 (the incomplete-gamma L the unfolding reproduces); arXiv:1603.03056; arXiv:2202.09542; Diamantis et al. (RMS 2020); Brown–Fonseca (arXiv:2508.04844). Frame the contribution as the finite, regulator-free Hurwitz-kernel *representation* unifying periods and L -values.

Paste-ready prompt for a Claude Code instance

Write a ~5-page note from this spine. Define-and-verify, not derive-and-discover.

1. Open with the construction question: how to build a parabolic cocycle for a modular form from a generic finite base point. Recall Brown (arXiv:1710.07912) in one paragraph; state the gate $c_f(0) = 0$.

2. Present the Hurwitz T -kernel $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)}\zeta(s-(k-1), \tau+1)$ as a definition. Verify $\delta_T \zeta(1-s, \tau+1) = -(\tau+1)^{s-1}$ and that the parabolic obstruction vanishes iff $c_f(0) = 0$. Footnote the five-step provenance; frame step 5 as revealing the entire-in- x , meromorphic-in- s object, not as continuing a discrete identity.
3. Period polynomials on $S_k^!$ from the two-segment functional.
4. L -function by swapping the period monomial for τ^{s-1} : entire on $S_k^!$, expected poles on $M_k^!$. $\tau_0 = i$ recovers BFK with $\Gamma(s, 2\pi n)$, $i^k \Gamma(k-s, 2\pi n)$; $b=1$ is the split height, not the s -center. State all- s as a result (coefficient-by-coefficient match to the incomplete-gamma series), not a sketch.
5. Encapsulation: weak/regular cusps (fully canonical) and interior poles (canonical given a contour; cross-wall jumps are f 's residues times the finite kernel value). State the symmetric winding prescription explicitly.
6. Relate to prior work: BFK (2014), arXiv:1809.02938, 1603.03056, 2202.09542, Diamantis et al. (RMS 2020), Brown–Fonseca (arXiv:2508.04844). Contribution is the finite regulator-free Hurwitz-kernel representation, not new objects.

Tone: canonical/certified, never “correct”; parsimonious novelty language; no claim broader than the simply-connected case without the winding clause.